

Get Ready: Setup Instructions & Workshop Details

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Applied Timeseries in R
7 Feb 2026
09:30 – 16:30
WB – 2003

Certificate in Engagement with Practice in Finance (CEPF)
Durham University Business School



Action Required

GitHub Setup

Workshop Information

Overview

Programme

Learning Outcomes

Action Required

Before the Workshop: GitHub Setup

All students must have a GitHub account and, optionally, start verification for the **GitHub Education as a student** in order to access frontier AI models for free through GitHub Copilot.

What you need to do

- 1 Create a GitHub account at <https://github.com>.
- 2 Choose a professional username (see right).
- 3 Sign up using your `durham.ac.uk` email.
- 4 Apply for the **GitHub Education as a student** (<https://github.com/settings/education/benefits>).
- 5 **Record your login details:** GitHub username, password, etc.
- 6 Bring your working login to the workshop (you should be able to sign in on your own laptop).

Username format (examples)

- `firstname-lastname`
- `firstname-lastname-yy`
- `firstinitial-lastname`

Key rules / tips

- Use only your `@durham.ac.uk` address to register/verify.
- Keep usernames professional; avoid nicknames and numbers where possible.

Workshop Information

About the Masterclass:

This one-day masterclass provides an intensive, hands-on introduction to applied time series econometrics in R.

The workshop is designed to help you:

- Develop practical financial modelling skills using time series data
- Strengthen empirical techniques relevant for dissertation work
- Prepare for Financial Risk Management in Term 2
- Gain confidence implementing econometric models in R

Highly recommended for students planning to work with financial, macroeconomic, or risk-related data.

Logistics

Date: Saturday 7th February 2026

Time: 09:30 – 16:30

Location: WB – 2003

Instructor

Dr Christian Engels

Econometrician with expertise in time series analysis, financial econometrics, and empirical modelling.

Time	Unit	Topic
09:30 – 10:00	0	Setup: GitHub, Codespaces, R environment
10:00 – 10:45	1	Time Series & Tidyverse Foundations
10:45 – 11:00		<i>Break</i>
11:00 – 12:00	2	Random Walks, Stationarity & Unit Root Tests
12:00 – 13:00		<i>Lunch</i>
13:00 – 14:00	3	ARMA/ARIMA Models & Forecasting
14:00 – 14:15		<i>Break</i>
14:15 – 15:15	4	Volatility Models (ARCH/GARCH)
15:15 – 15:30		<i>Break</i>
15:30 – 16:15	5	Portfolio Optimisation & Risk Management
16:15 – 16:30	6	Wrap-up & Resources

You will be able to:

- 1 Download, wrangle and visualise financial time series in R
- 2 Test for stationarity and interpret ACF/PACF plots
- 3 Fit and forecast ARIMA models for the conditional mean
- 4 Specify, estimate and compare GARCH models for volatility
- 5 Construct an optimised portfolio and backtest its Value at Risk

Each unit is delivered as a pair:

- **Slide deck** — theory, equations, key ideas
- **Companion R script** — code you run alongside (named `Unit-N-Topic.R`, e.g. `Unit-1-TimeSeries-Foundations.R`)

Hands-On Workshop

You will write and run R code throughout the day. Experiment, break things, ask questions!

See you at the workshop!